FI360 TYRE MODULE
SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Fleet Intelligence 360 (FI360) — Version 1.0
Fleet Associates Africa Ltd (FAA)
9 August 2026

Document Control

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Parent	FI360 Module Development Standard v1.0
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Module	FI360 Tyre Management
Owner	Fleet Associates Africa Ltd
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1. Executive Summary

FI360 Tyre is a standalone enterprise tyre-management module designed to control the complete tyre lifecycle from receipt and registration through inspection, storage, fitment, rotation, repair, retreading, removal, costing, performance analysis and final disposition. It shall operate independently for a customer that buys only the Tyre module, while exposing governed interfaces that allow later connection to FI360 Workshop, Fleet, Parts, Telematics and Analytics.

FI360 TYRE MODULE

User / UI → Tyre Module → FI360 Core → Data / Events

2. Business Objectives

- Create a single controlled digital record for every tyre.
- Improve tyre availability, utilisation and lifecycle visibility.
- Reduce premature tyre loss and unexplained tyre costs.
- Control tyre purchasing, receipt, storage and issue.
- Track tyre fitment by vehicle, axle and position.
- Capture inspection, tread/depth and condition history.
- Control tyre repair, retreading and disposition.
- Provide tyre cost-per-kilometre and lifecycle KPIs.
- Create an auditable history of tyre movements and decisions.

FI360 Tyre principle: every tyre has one identity, one traceable lifecycle and one auditable history.

- Allow future integration with Workshop, Fleet and Telematics without redesigning the Tyre module.

3. Scope

3.1 In Scope

- Tyre master register
- Tyre receiving
- Tyre inspection
- Tyre storage
- Tyre issue
- Tyre fitment
- Tyre removal
- Tyre rotation
- Tyre repair
- Retreading lifecycle
- Tread/depth recording
- Tyre pressure recording
- Damage/condition classification
- Tyre movement history
- Tyre costing
- Tyre performance KPIs
- Alerts and exceptions
- Tyre reports
- API/event integration
- Audit trail

3.2 Out of Scope for Initial Release

- Full workshop work-order management
- Full vehicle fleet master management
- Full parts inventory management
- Fuel management
- Telematics device management
- Enterprise financial accounting

These may be integrated later through FI360 Core contracts.

4. Users and Roles

Role	Primary Functions
Tyre Administrator	Configuration, tyre master, permissions and controls
Tyre Manager	Policy, approvals, KPI and performance management
Tyre Supervisor	Daily operational control, inspections, fitment/removal approval
Tyre Technician	Fitment, removal, inspection and technical records
Stores Officer	Receipt, storage, issue and stock control
Workshop User	Tyre-related transactions from approved workshop workflows

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Fleet Manager	Tyre performance and vehicle-level visibility
Finance/Cost Analyst	Tyre expenditure and lifecycle cost analysis
Auditor	Controlled read-only review
System Integration	Machine-to-machine API/event access

5. Core Tyre Concepts

Entity	Purpose
Tyre	Unique physical tyre record
Tyre Brand/Model	Manufacturer/model reference
Tyre Specification	Size, construction, load/speed rating and technical attributes
Tyre Position	Vehicle wheel/axle position
Fitment	Tyre-to-asset position transaction
Inspection	Condition assessment at a point in time
Tread Reading	Measured tread depth
Pressure Reading	Measured inflation pressure
Movement	Controlled tyre movement between locations/states
Repair	Tyre repair event
Retread	Retreading lifecycle record
Removal	Removal transaction and reason
Disposition	Final action such as scrap, sale or return
Cost Transaction	Acquisition, repair, retread or other lifecycle cost

6. Tyre Identification

Every physical tyre must have a unique FI360 Tyre ID. Manufacturer serial number, DOT/production marking and other identifiers are retained as attributes.

Identifier	Example
FI360 Tyre ID	FI360-TYR-000001
Manufacturer Serial	Manufacturer-provided serial
DOT / Production Code	Captured where applicable
Barcode/QR	Machine-readable FI360 identifier
Brand	Manufacturer brand
Pattern/Model	Manufacturer pattern/model
Size	Example: 315/80R22.5
Position Type	Steer / Drive / Trailer / Other

7. Tyre Registration

The system shall support individual registration and controlled bulk import.

REGISTRATION

Receive → Inspect → Register → Activate

ID	Requirement
FR-TYR-001	Create a unique FI360 Tyre ID.
FR-TYR-002	Capture required tyre specification data.

FI360 Tyre principle: every tyre has one identity, one traceable lifecycle and one auditable history.

FR-TYR-003	Prevent duplicate manufacturer identifiers where configured.
FR-TYR-004	Assign initial lifecycle status.
FR-TYR-005	Record source, supplier and receipt reference where applicable.
FR-TYR-006	Audit creation and material changes.

8. Receiving and Stores

- Receive tyres against an approved order/receiving reference.
- Record quantity, supplier, specification and condition.
- Record storage location.
- Support barcode/QR scanning where available.
- Flag damaged or incorrect tyres before activation.
- Maintain stock status.
- Support controlled transfers between storage locations.

FR	Requirement
FR-TYR-010	Record tyre receipt.
FR-TYR-011	Assign storage location.
FR-TYR-012	Record receiving inspection.
FR-TYR-013	Prevent issue of tyres not in an issueable status.
FR-TYR-014	Maintain movement history.

9. Fitment

Fitment connects a tyre to a canonical FI360 Asset ID and defined vehicle position. The Tyre module references the Fleet module's canonical asset identifier when Fleet is installed.

FITMENT

Select Tyre → Validate → Fit → Post Event

ID	Requirement
FR-TYR-020	Select an eligible tyre.
FR-TYR-021	Select a valid FI360 Asset ID.
FR-TYR-022	Select a valid wheel/axle position.
FR-TYR-023	Validate tyre specification against position rules.
FR-TYR-024	Record fitment date, mileage/hours and user.
FR-TYR-025	Close/supersede previous fitment at the same position according to policy.
FR-TYR-026	Publish TYRE_FITTED event after successful posting.

10. Tyre Inspection

Inspections are time-stamped condition assessments and must preserve historical readings rather than overwriting previous inspections.

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INSPECTION

Inspect → Record → Assess → Action

Inspection Item	Examples
Tread	Depth by tread point/position
Pressure	Measured pressure and unit
Visual	Cuts, cracks, bulges, penetration, uneven wear
Condition	Serviceable / monitor / remove
Temperature	Where appropriate
Photo	Evidence attachment
Technician	Person performing inspection

11. Rotation

- Select eligible fitted tyres.
- Validate vehicle/position rules.
- Record old and new position.
- Record date and mileage/hours.
- Record reason.
- Update lifecycle history.
- Publish TYRE_ROTATED event where configured.

12. Removal

ID	Requirement
FR-TYR-030	Record tyre removal from asset/position.
FR-TYR-031	Require removal reason.
FR-TYR-032	Capture removal mileage/hours.
FR-TYR-033	Capture condition and remaining tread where applicable.
FR-TYR-034	Move tyre to an appropriate post-removal status/location.
FR-TYR-035	Publish TYRE_REMOVED event.

13. Repair and Retread

- Record repair type, date, supplier/technician and cost.
- Track repair count.
- Record retread cycle number.
- Capture retread supplier and cost.
- Track casing identity.
- Prevent invalid lifecycle transitions.
- Maintain total lifecycle cost.

Lifecycle	Example Status
New casing	NEW
In storage	IN_STOCK
Fitted	FITTED

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Removed	REMOVED
Under repair	REPAIR
Retread processing	RETREAD
Ready for service	SERVICEABLE
Retired	RETIRED
Scrapped	SCRAPPED

14. Tyre Lifecycle

LIFECYCLE

New → Fitted → In Service → Removed / Retired

The transition matrix shall be configurable within approved limits; invalid transitions must be rejected server-side.

15. Tyre Costing

- Acquisition cost.
- Transport/receiving cost where captured.
- Repair cost.
- Retread cost.
- Disposal or residual value.
- Total lifecycle cost.
- Cost per kilometre/hour where mileage/hours are available.

KPI	Definition
Cost per km	Relevant lifecycle cost ÷ tyre kilometres
Cost per hour	Relevant lifecycle cost ÷ operating hours
Average life	Average distance/hours from fitment to removal
Retread yield	Retreaded casings returning to service ÷ casings sent for retread
Premature removal rate	Tyres removed before configured target life ÷ tyres removed
Stock ageing	Tyres in stock grouped by age bands

16. Alerts and Exceptions

Alert	Trigger Example
Low stock	Available stock below configured minimum
Abnormal wear	Inspection indicates configured wear pattern
Low tread	Reading reaches configured warning threshold
Pressure exception	Reading outside configured tolerance
Tyre ageing	Tyre exceeds configured age threshold
Repeated failure	Repeated repair/removal within configured period
High cost	Lifecycle cost exceeds configured threshold
Missing inspection	Required inspection overdue

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17. Dashboards

Dashboard	Core Indicators
Tyre Executive	Total tyres, active tyres, cost/km, lifecycle cost, premature removals
Tyre Operations	Stock, fitments, removals, inspections, overdue actions
Tyre Performance	Life, wear, failure, retread and cost trends
Vehicle Tyre View	Tyres by vehicle/position and current status
Stores View	Stock by size, brand, location and age
Management Exceptions	Alerts, overdue inspections and abnormal events

18. Reports

- Tyre register.
- Tyre stock report.
- Tyre fitment history.
- Tyre removal report.
- Inspection report.
- Tread-depth trend.
- Pressure compliance report.
- Tyre movement history.
- Repair history.
- Retread history.
- Tyre cost report.
- Cost-per-kilometre report.
- Premature failure report.
- Tyre position analysis.
- Supplier/brand/model performance.
- Scrap/disposition report.
- Audit report.

19. Search and Filtering

- Tyre ID.
- Manufacturer serial.
- Brand/model.
- Size.
- Vehicle/Asset ID.
- Registration number where supplied by Fleet.
- Position.
- Status.
- Location.
- Supplier.
- Date range.
- Tread range.
- Cost range.
- Alert status.

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20. Permissions

Permission	Example
tyre.read	View tyre records
tyre.create	Register tyre
tyre.receive	Receive stock
tyre.fit	Fit tyre
tyre.remove	Remove tyre
tyre.inspect	Perform inspection
tyre.rotate	Rotate tyres
tyre.repair	Record repair
tyre.retread	Manage retread
tyre.cost	View/manage cost
tyre.approve	Approve controlled transactions
tyre.admin	Module administration
tyre.export	Export approved reports

21. APIs

API Area	Purpose
Tyre Master API	Create/read/update controlled tyre data
Fitment API	Post/read fitment records
Inspection API	Post/read inspection records
Movement API	Record tyre movements
Lifecycle API	Controlled status transitions
Cost API	Lifecycle cost records
Analytics API	Approved KPI/query access
Integration API	Controlled external/module integration

22. Events

Event	When Published
TYRE_REGISTERED	Tyre successfully registered
TYRE_RECEIVED	Tyre receipt posted
TYRE_FITTED	Fitment posted
TYRE_REMOVED	Removal posted
TYRE_ROTATED	Rotation posted
TYRE_INSPECTED	Inspection posted
TYRE_REPAIRED	Repair posted
TYRE_RETREADED	Retread cycle completed
TYRE_RETIRED	Tyre retired
TYRE_SCRAPPED	Tyre scrapped

23. Integration with Other FI360 Modules

INTEGRATION

Tyre → API / Event → FI360 Core → Workshop / Fleet

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Module	Integration Purpose
Fleet	Asset ID, vehicle context, mileage/hours
Workshop	Work orders and tyre-related maintenance activities
Parts	Tyre/related stock or consumable relationships where appropriate
Telematics	Mileage/hours/vehicle usage signals
Fuel	Potential correlation with vehicle operating cost
Analytics	Cross-module KPI and TCO analysis

A customer purchasing only FI360 Tyre must not need these modules. When another module is later purchased, the existing tenant, Asset IDs, users and integration contracts enable connection.

24. Recommended Technology Stack

The recommended FI360 Tyre technology stack is enterprise-oriented, strongly typed and suitable for deployment from a small single-site customer to a large multi-site, multi-tenant enterprise. The objective is a coherent stack with a small number of primary technologies.

Layer	Recommended Technology	Why
Primary application language	TypeScript	Strong typing, maintainability and shared language across web/API/mobile
Web frontend	React + TypeScript	Mature ecosystem, reusable components and enterprise UI capability
Backend/API	NestJS + TypeScript	Modular architecture, dependency injection, validation and maintainable services
Database	PostgreSQL	Strong relational integrity, transactions, indexing and enterprise maturity
Data access	Prisma or TypeORM — standardize on one	Typed access and migration support
Cache / jobs	Redis	Caching and background processing where required
Messaging	Kafka or RabbitMQ, selected by scale/use case	Reliable asynchronous module integration
Mobile	React Native + TypeScript	Efficient mobile development using the primary application language
Analytics / AI	Python	Specialist analytics, forecasting, optimization and machine learning
API contract	OpenAPI	Standardized, machine-readable API definition
Authentication	OIDC/OAuth 2.0 compatible provider	Enterprise SSO and identity integration
Packaging	Docker	Repeatable deployment
Orchestration	Kubernetes when justified by scale	Container orchestration and horizontal scaling
CI/CD	Git-based CI/CD	Automated build, test and deployment
Observability	OpenTelemetry-compatible stack	Logs, metrics and distributed tracing

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25. Primary Language Decision

TypeScript should be the primary FI360 application-development language. Using TypeScript for both React frontend and NestJS backend reduces language switching, enables shared types/contracts and supports a consistent engineering standard across FI360 modules.

25.1 Python Usage

Python should be a specialist language for analytics, forecasting, optimization and AI/ML services. It should not replace the core transactional TypeScript backend simply because Python is popular in AI.

25.2 Why This Stack Is Suitable at Different Levels

Scale	Recommended Approach
Small customer	React + TypeScript, NestJS + TypeScript, PostgreSQL; deploy as a compact application
Growing customer	Separate services as load requires; add Redis/background jobs
Multi-site enterprise	Containerized services, managed PostgreSQL, messaging, observability and stronger HA
Large FI360 platform	Kubernetes where justified, independently scalable services, event-driven integrations, dedicated analytics/AI services

The architecture should scale by separating bottlenecks when necessary rather than starting with unnecessary microservice complexity.

26. Technology Architecture Requirements

- Use strict TypeScript settings.
- Maintain API contracts and shared schemas.
- Keep business rules testable independently of the UI.
- Version database migrations.
- Never expose the production database directly to browsers.
- Store secrets outside source code.
- Expose health/readiness checks.
- Include correlation IDs in logs.
- Version Core APIs/events.
- Standardize technology choices across modules unless an approved exception exists.

27. Non-Functional Requirements

Area	Target Principle
Availability	Defined by deployment tier; critical services designed for graceful recovery
Security	Tenant isolation and server-side authorization mandatory
Performance	Predictable response for critical operational actions
Scalability	Support small deployments through multi-site enterprise
Data integrity	Transactional consistency for critical tyre operations
Auditability	Material changes traceable to actor/service and time
Recoverability	Backups, restore testing and defined RPO/RTO

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Maintainability	Modular, documented and testable code
Observability	Logs, metrics, traces and health checks
Compatibility	Versioned APIs/events and controlled migrations

28. Initial Data Model

Entity	Key Data
Tyre	ID, serial, brand, model, size, type, manufacture date, status
TyreSpecification	Size, construction, load/speed rating, tread pattern
TyreLocation	Site, store, bay/bin, status
TyreFitment	Tyre, asset, position, date, mileage/hours, user
TyreInspection	Tyre, date, inspector, condition, tread, pressure, findings
TyreMovement	From, to, date, reason, actor
TyreRepair	Date, type, supplier, cost, result
TyreRetread	Cycle, supplier, date, cost, result
TyreCost	Type, amount, currency, date, reference
TyreDisposition	Reason, date, value, destination
TyreEventReference	Event ID, event type, correlation ID

29. Business Rules

- Only an eligible tyre can be fitted.
- A tyre cannot be actively fitted to two positions at the same time.
- A vehicle position cannot have two active tyres unless explicitly configured.
- Removal closes the active fitment relationship.
- Historical transactions must not be silently overwritten.
- Closed/approved transactions require controlled correction mechanisms.
- Retread count cannot decrease through ordinary transactions.
- Scrapped/retired tyres cannot return to service without an approved exceptional workflow.
- Tenant ownership must be preserved.
- Material lifecycle transitions must be auditable.

30. Acceptance Criteria

Area	Acceptance Condition
Registration	Unique tyre can be registered and searched
Stock	Received tyre appears in correct stock/location status
Fitment	Eligible tyre can be fitted to valid Asset ID and position
Inspection	Inspection records preserve history
Removal	Removal closes active fitment and records reason
Rotation	Rotation preserves before/after position history
Repair	Repair cost and event are recorded
Retread	Retread cycle and cost are tracked
Lifecycle	Invalid transitions are rejected
Reporting	Core operational and management reports work
Security	Unauthorized users cannot perform protected actions
Audit	Material actions are traceable
Integration	API/event contracts can connect future modules
Standalone	Tyre module remains functional without optional modules

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31. Testing Strategy

1. Unit-test business rules and calculations.
2. Integration-test database and FI360 Core interfaces.
3. Test tenant isolation and authorization.
4. Test API validation and error handling.
5. Test event publication and duplicate delivery.
6. Test critical end-to-end workflows.
7. Conduct performance tests on high-volume operations.
8. Complete UAT using realistic customer scenarios.
9. Execute regression testing before release.

32. Initial Development Backlog

Epic	Priority	Key Deliverables
Foundation	P0	Tenant/Core integration, authentication, permissions, module setup
Tyre Master	P0	Tyre register, specifications, identifiers and search
Stores	P0	Receipt, locations, stock and movements
Fitment	P0	Asset/position fitment and removal
Inspection	P0	Inspection, tread, pressure and condition
Lifecycle	P0	Rotation, repair, retread, retirement and scrap
Costing	P1	Lifecycle cost and cost/km
Alerts	P1	Thresholds and exceptions
Dashboards	P1	Operational and management dashboards
Reports	P1	Standard reports and exports
Integration	P1	API/event contracts
Advanced Analytics	P2	Predictive wear, failure and optimization

33. Suggested Development Phases

Phase	Scope
Phase 1	Foundation + Tyre Master + Stores
Phase 2	Fitment + Removal + Inspection
Phase 3	Rotation + Repair + Retread + Lifecycle
Phase 4	Costing + Alerts + Reports + Dashboards
Phase 5	API/events + Workshop/Fleet integration
Phase 6	Advanced analytics and predictive tyre management

34. Recommended First MVP

The first commercially viable release should focus on the workflows that create immediate operational value rather than attempting every advanced feature at once.

- Tyre registration.

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- Tyre stock and location.
- Fitment/removal.
- Inspection and tread recording.
- Tyre history.
- Basic costing.
- Basic alerts.
- Core operational dashboard.
- Essential reports.
- FI360 Core integration.
- API/event foundation.

35. Future Advanced Capabilities

- Predictive tyre replacement.
- AI-assisted abnormal wear detection.
- Image-assisted tyre condition assessment.
- Automatic mileage-based inspection triggers.
- Telematics-driven lifecycle calculations.
- Supplier scorecards.
- Retread optimization.
- Vehicle-level tyre TCO.
- Tyre failure prediction.
- Fleet-wide tyre benchmarking.

36. Definition of Done for FI360 Tyre v1.0

- Approved SRS.
- Approved UI/UX.
- Approved data model.
- Core integration complete.
- Role/permission matrix implemented.
- P0 workflows implemented.
- Unit/integration/system tests passed.
- Security and tenant-isolation tests passed.
- UAT completed.
- API/event contracts documented.
- Deployment and rollback procedures documented.
- Monitoring and audit enabled.
- User and administrator documentation completed.

37. Next Document

DOCUMENT 05 — FI360 TYRE MODULE FUNCTIONAL ARCHITECTURE & DATA MODEL will translate this SRS into the detailed module architecture, entity-relationship model, database tables, field-level definitions, state-transition matrix, API objects and screen/module map needed by the development team.

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